69. Internal Voltage Regulator

69.1 Overview

The MCU supports two power supply providing for core voltage (VDD):

- Switching regulator (DCDC)
- External power supply from external power device (External VDD)

Table 69.1 shows the specifications of internal voltage regulator.

DCDC or External VDD is selected with the OFS setting.

- When OFS2.DCDCEN is 1, DCDC is selected (DCDC mode)
- When OFS2.DCDCEN is 0, External VDD is selected (External VDD mode)

Table 69.1 Power supply selection

OFS setting (OFS2.DCDCEN)	Power mode
1	DCDC mode
0	External VDD mode

69.2 Operation

69.2.1 DCDC Mode

Table 69.2 lists the DCDC mode pin settings, and Figure 69.1 shows the DCDC mode settings.

In DCDC mode, VDD is supplied from VCL through VLO output and an external inductor and capacitor.

Table 69.2 DCDC mode pin settings

Pins	Setting descriptions
All VCC_n and VCC2_n pins	• Connect each pin to the system power supply. • Connect each pin to the same numbered VSS_n through a 0.1-μF multilayer ceramic capacitor. Place the capacitor close to the pin.
VCC_DCDC pins	• Connect the pin to the system power supply. • Connect the pin to VSS_DCDC through a 22-μF and 0.1-μF multilayer ceramic capacitor in parallel. Place the capacitor close to the pin.
VCLn pins	• Connect each pin to the same numbered VSSn through a 0.22-μF multilayer ceramic capacitor. Place the capacitor close to the pin. • Connect each pin to an external inductor and capacitor. Place the inductor and capacitor close to the pin. See Figure 69.1.
All VLO pins	Connect each pin to an external inductor and capacitor. Place the 2.2-μH inductor and 47-μF capacitor close to the pin. This capacitor must be grounded to VSS_DCDC

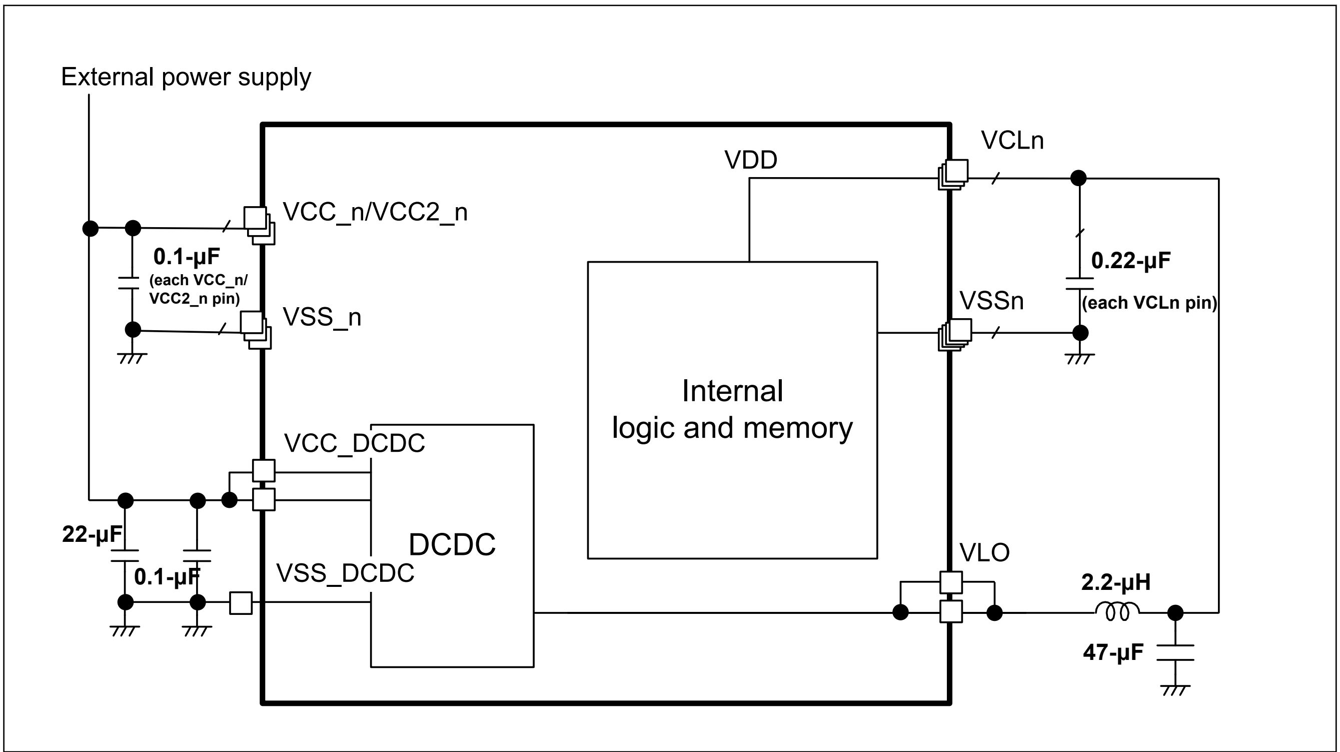

Figure 69.1 DCDC mode setting

69.2.2 External VDD Mode

Table 69.3 lists the External VDD mode pin settings, and Figure 69.2 shows the External VDD mode settings.

VDD is supplied from VCL pins.

Note: Software Standby mode, Deep Software Standby mode 1, 2, 3, Battery Backup Function and Voltage Scaling Control are not supported in this mode.

Table 69.3 Setting descriptions for External VDD mode

Pins	Setting descriptions
All VCC_n and VCC2_n pins	• Connect each pin to the system power supply device for VCC. • Connect each pin to the same numbered VSS_n through a 0.1-µF multilayer ceramic capacitor. Place the capacitor close to the pin.
VCC_DCDC pins	• Connect the pin to VSS_DCDC through 0.1-µF multilayer ceramic capacitor. Place the capacitor close to the pin.
VCLn pins	• Connect each pin to the system power supply device for VDD. • Connect each pin to the same numbered VSSn through a 0.22-µF multilayer ceramic capacitor. Place the capacitor close to the pin.
All VLO pins	Keep pin open

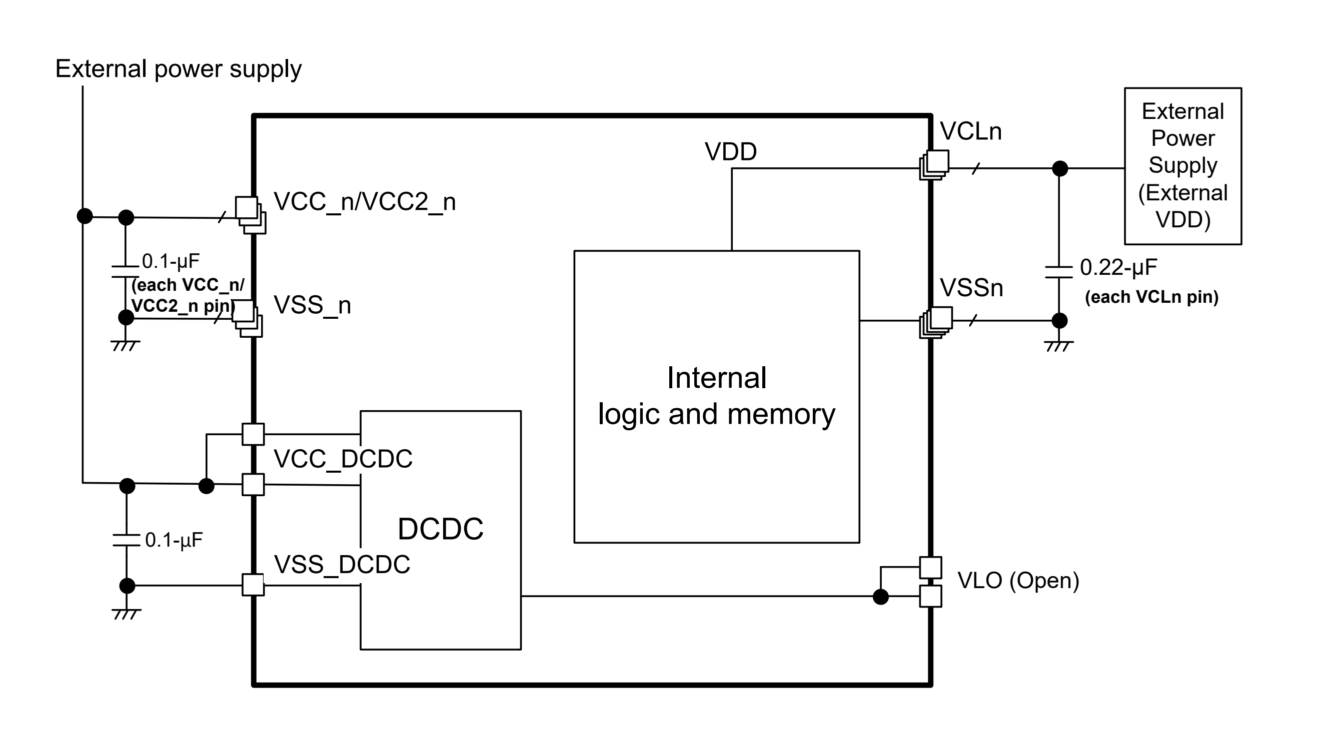

Figure 69.2 External VDD mode setting

69.3 Usage Notes

In DCDC mode, to minimize power loss and maximize efficiency, Renesas recommends using a 2.2-µH inductor with a DC resistance of 100 mΩ or less. Short VCC_n and VCC_DCDC.

In external VDD mode, the voltage of VDD should always be below the voltage of VCC including the power-on and power-off sequence. For details, see section 70, Electrical Characteristics.